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Development of a novel EEG rating scale for head injury using dichotomous variables

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Summary We developed a new EEG rating scale for electrographic assessment of head injured patients. Phenomena present in posttraumatic EEG were scored as dichotomous variables (present or absent). These phenomena included background activity (alpha, beta, theta, delta), sleep spindles, focal abnormalities, reactivity and variability, epileptiform activity, and specific comatose patterns. Each variable was weighted according to its perceived prognostic value: i.e., normal alpha 10, flat EEG −10, spindles 4, etc. Combinations of possible scores ranged from +23 to −10.

Fifty-seven EEGs from different head injured patients were independently and retrospectively analyzed by two investigators. There was a high correlation for intra- ($r = 0.95$) and inter- ($r = 0.85$) observer rating using the dichotomous test. When patients with scores over 15 (i.e., with reactive alpha) and patients with scores of −10 (i.e., ECI records) were excluded, the intra-rater and inter-rater correlations were still high (0.81 and 0.76, respectively). There was a high correlation between Glasgow outcome score at discharge and the dichotomous EEG score.

This EEG scale scores most major categories of EEG activity, utilizes a multipoint scale for correlation purposes, and allows data to be analyzed in sub-categories (i.e., spindles in coma). The separate weighting score allows for refinement of the scale after data collection (i.e., to fit prospective outcome).

We feel that this scale is reproducible and valid, and may be applicable to other patient groups with severely altered EEGs.

Key words: EEG; Scale; Head injury; Dichotomous variables

Head trauma in the 1990s has become known as the "silent epidemic," causing half a million admissions per year in the United States and precipitating lifelong disability in 90,000 new patients each year (Goldstein 1990). Sixty percent of trauma deaths occur among patients with head injury (Gennarelli 1989). In addition, head injuries occur among the young and employable, compounding the social, personal, and medical tragedy (Gennarelli 1989). Predicting the ultimate outcome is important in providing rational health care to this large and increasingly important group of patients. Continued development of tools to improve on the clinical examination is necessary.

As part of a study of neurophysiological changes in major head injuries, we developed a rating scale for the EEG based on the presence or absence of easily identifiable features. Our goal was to develop a reliable and valid tool to score EEG in a prospective fashion to compare with other indices of neurophysiological function.

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Varyingly successful attempts at scoring the comatose EEG have been made. Hockaday et al. (1965) proposed a scale based on background frequency. Progressively worse EEGs were scored on the basis of slowing with a progression through theta to delta, invariant delta, and electrocerebral inactivity. The rich variety of EEG abnormalities in traumatic and other comas are lost in this scheme. In addition some comas with poor prognosis such as alpha coma and theta coma are either not scorable or are placed in a misleading grade. Similar indices have been suggested by Prior (1973) and Markand (1984). Hughes et al.'s (1976) scale scored the variety and quantity of frequencies present. The subjective nature of determining the percentage of each frequency present makes this difficult to use. Patients with many frequencies, despite their otherwise poor prognostic pattern, are scored high with this system.

Most recently Synek (1988a,b, 1990) has reported and refined a classification for the EEG in coma which has the merits of accentuating both the specific patterns and the general character of the EEG. Synek found a correlation between poorer grades on the Synek scale and a poor outcome, validating his observations.

We attempted to take lessons learned from these scales and apply them to patients with traumatic brain injury to develop a validated, reproducible and quantitative neurophysiological tool.

Methods

Portable and laboratory EEGs were performed according to American EEG Society Guidelines (1986) using Grass model 16 and Nihon-Kohden model 4418A EEG machines. Silver-silver chloride electrodes, 21 channels according to the 10–20 system, and standard referential and bipolar montages were used for a recording period of 30 min or more. Stimulation with a photic stimulator, sound, and pain was performed during the study. Studies were performed under the supervision of a registered technologist (REEGT).

For EEG variables, we selected those features which were found to have prognostic value in well studied series of traumatic and non-traumatic comatose patients (Chatrian and White 1961; Chatrian et al. 1963, 1964; Bergamasco et al. 1968; Bricolo and Turella 1973; Brenner et al. 1975; Lorenzoni 1975; Westmoreland et al. 1975; Hughes et al. 1976; Rumpl et al. 1979, 1983; Kuroiwa and Celesia 1980; Hansotia et al. 1981; Iragui and McCutcheon 1983; Synek and Synek 1987; Austin et al. 1988; Synek et al. 1988a,b, 1990; Dusser et al. 1989). We then developed a scale which included each of these features in a present or absent dichotomy, rather than as a qualitative measure. We then gave each feature a weighting from +10 (good prognostically) to −10 (poor prognostically) based on the usual significance of the wave form (Brenner et al. 1975; Westmoreland et al. 1975; Rumpl et al. 1983; Dusser et al. 1989; Synek 1990).

We reviewed all EEG records performed on patients admitted to the Shock Trauma Unit (STU) or Neurological Intensive Care Unit (NICU) of the Lehigh Valley Hospital Center over 3 years (1987–March 1990). During this time, an EEG was performed on all patients to investigate seizures, focal neurological abnormalities, CNS deteriorations, or suspected brain death. A total of 57 first EEGs were scored using this scale.

For comparison and validation against presently used EEG rating scales, we also scored these studies using coma scales proposed and studied by Hockaday et al. (1965), Hughes et al. (1976), and Synek (1988a,b, 1990). One electroencephalographer (ARG) scored each record twice (6 weeks apart) without reviewing the prior score or case record, while another (PB) scored the records once independently.

Data were entered in a computer database and analyzed statistically. Pearson's product moment correlation (r) was computed between the dichotomous scale on the one hand and, in turn, the Hockaday, Synek, and Hughes scales. A contingency coefficient measure of association (C) was computed between the Synek scale and the Hockaday scale.

TABLE I

Dichotomous rating scale for EEG.

		Weight
Background activity		
'Normal' alpha	Present Absent	+10
WAU/alpha coma	Present Absent	−4
Beta	Present Absent	+2
Theta	Present Absent	+1
"Theta coma"	Present Absent	−2
Delta (general/frontal)	Present Absent	−1
Delta (focal, non-frontal)	Present Absent	−1
Spindles (symmetric)	Present Absent	+4
Spindles (asym. or abn.)	Present Absent	−2
Symmetry, reactivity, variability		
Asymmetry (not posterior)	Present Absent	−2
Posterior suppression	Present Absent	−2
Reactivity	Present Absent	+3
Variability	Present Absent	+3
Additional patterns		
IRDA	Present Absent	0
Triphasic waves	Present Absent	0
ELAE	Present Absent	−3
Epileptiform activity	Present Absent	−1
Burst suppression	Present Absent	−5
Low voltage pattern	Present Absent	−6
ECI	Present Absent	−10
	Dichotomous score:	–

Corresponding intrarater (first versus second interpretation (ARG)) and interrater (ARG versus PB) association statistics were computed. Because of the number of significance tests conducted, we used a Bonferroni adjusted significance level of 0.005.

Charts were retrospectively reviewed and the patient outcome assessed from clinical notes at discharge. An assessment of Glasgow outcome scale (GOS) was then made (i.e., 1 = death, 2 = vegetative state, 3 = severely disabled, 4 = moderately disabled, 5 = good outcome (Jennett et al. 1976)). A comparison between GOS at discharge and first EEG dichotomous score was made using Pearson correlation coefficients.

Study tools

A scoring sheet was developed for the "Dichotomous rating scale" utilizing the features obtained (see Table I). Variables were grouped under 3 major categories: (a) background activity, (b) symmetry, reactivity and variability, and (c) specific abnormal patterns.

The presence or absence of each feature was rated independently of other features. Where features overlapped, each was scored independently. For example, patients with "theta coma" (Synek 1988a,b) were scored present for theta (i.e., +1) and for theta "coma" pattern (i.e., −2). The presence of a pattern was scored as long as it was clearly visualized at some point in the record. No attempt was made to score the "severity," "frequency" or "amount" of such patterns. Amplitude was not scored except under the category of "low voltage pattern," as suggested by Hughes et al. (1976).

The features scored were defined as follows:

(1) "Normal" alpha: reactive background alpha with normal posterior head distribution and evidence of blocking with eye opening [1].
(2) Widespread anterior unresponsive alpha (WAU)/ alpha coma: widespread, usually anteriorly distributed alpha which does not have the response or field seen with normal alpha activity [2] (Chatrian et al. 1964; Westmoreland et al. 1975; Bauer et al. 1982; Zaret 1985).
(3) Beta: fast activity of 14 Hz or greater [3] (Stockard et al. 1975).
(4) Theta: background activity in the 4–7 Hz range.
(5) "Theta coma": comatose pattern with theta range activity as the major background activity, often intermixed with brief suppressions (Synek and Synek 1984, 1987).
(6) Delta (generalized or frontal): 1–3 Hz background activity which is generalized or frontally predominant, excluding focal delta not in the frontal regions (Stockard et al. 1975).
(7) Delta (focal, non-frontal): focal 1–3 Hz activity in any region other than frontal.
(8) Spindles (symmetric): 12–14 Hz well organized bursts of sinusoidal activity lasting seconds, often maximal in central regions (Chatrian and White 1961; Chatrian et al. 1963; Bergamasco et al. 1968; Hansotia et al. 1981; Rumpl et al. 1983).
(9) Spindles (asymmetric or abnormal): lower frequency (6–11 Hz) or distorted morphology bursts suggestive of spindles or asymmetric spindle activity (Silverman 1963; Rumpl 1981).
(10) Asymmetry: persistent and easily visualized non-artifactual difference in amplitude or frequency of background activity (Stockard et al. 1975).
(11) Posterior suppression: suppression of background activity involving posterior areas of the head with relative sparing of other areas of cortical activity (Stockard et al. 1975).
(12) Reactivity: reproducible response to external stimulation, with a change in background amplitude or frequency or abnormal patterns (Markand, 1984).
(13) Variability: variation in the background activity or specific patterns of activity independent of the response to external stimulation (Fischgold and Mathis, 1959; Bricolo and Turella 1972; Dusser et al. 1989).
(14) IRDA: intermittent, rhythmic delta activity occurring in bursts in either frontal or occipital regions (Bricolo and Turella 1973; Markand 1984).
(15) Triphasic waves: triphasic morphology waves with fronto-occipital delay or occipito-frontal delay, often with periodicity (Stockard, 1975; Kuroiwa and Celesia 1980).
(16) Episodic low amplitude events (ELAEs): brief (1–5 sec) epochs of suppression of background activity with variable background activity which is not dominated by a burst suppression pattern (Brenner et al. 1975; Rae-Grant and Barbour 1991).
(17) Epileptiform activity: spikes, stereotyped sharp waves or electrographic seizure patterns during the recording (Synek 1988a,b).
(18) Burst suppression: an alternating pattern of bursts of high voltage spikes, sharp and slow waves with intervening intervals of suppression of background activity (Kuroiwa and Celesia 1980; Sananman 1983).
(19) Low voltage pattern: background activity throughout record consisting of low voltage activity no greater than 10 μV amplitude measured on ear referential electrodes (Hockaday et al. 1965; Hughes et al. 1976; Synek 1988a,b).
(20) ECI: electrocerebral inactivity throughout record in accordance with American EEG Society Guidelines (1986).

The weighting of each subcategory was then added for an overall "dichotomous score."

Each EEG was also scored using previously defined coma rating scales shown in Table II (Hockaday et al. 1965; Hughes et al. 1976; Synek 1988a,b 1990). Where a record could not be scored it was rated "not scorable" for that scale. Since the Synek scale uses categorical subscales (i.e., 4A, 3B, etc.), for the purposes of cross-correlation we assigned fractional values for these scales (i.e., 4A = 4.0, 4E = 4.8, etc.).

Results

Demographic results were tabulated for the group of patients (Table III). Fifty-seven patients were scored using these scales. The average age was 37.5 ± 2.8 years. There were 38 males and 19 females. Forty-seven patients were involved in motor vehicle accidents, 7 in falls, 1 in a skiing accident, 1 with a gunshot wound to

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[1] "Normal" does not imply absence of other features such as theta or delta activity.

[2] Again presence of other patterns accepted.

[3] Excluding activity with characteristics of typical or abnormal spindles.

the head, and 1 with a crush injury of the head. Sedative, paralyzing, anesthetic and narcotic medications were frequently used in this group of patients. Nine were on paralyzing agents at the time of the study, 26 on morphine, 25 on benzodiazepines, 9 on anticonvulsants, and 3 on haloperidol. Five patients had had seizures. The time from trauma to first EEG varied from 1 to 60 days, the average being 9.9 ± 12.6 days. CT scans were abnormal in 36/57 patients (63%). Abnormalities included subarachnoid hemorrhage (in 14 patients), contusions (13), intraventricular hemorrhages (8), subdural hematomas (6), epidural hematomas (3), intracerebral hemorrhages (4), and stroke (2). These were often found in combination.

Twelve had a "good" outcome (either normal, or confused but talking and ambulatory at discharge), 8 had moderate disability, 17 were severely disabled (awake, little speech or severe motor disorder), 6 were in a vegetative state, and 14 died.

TABLE II

Rating scales for EEG in coma.

Hockaday (1965)

Grade 1: background alpha with/without scattered theta.
Grade 2: theta intermixed with some alpha or some delta.
Grade 3: continuous polymorphic delta with little faster frequencies. EEG variability and reactivity.
Grade 4: invariant delta activity of relatively small amplitude (< 100 μV) unresponsive to any stimulation or 'suppression burst' pattern.
Grade 5: near flat or ECI.

Hughes (1976)

'EEG INDEX.' All EEGs run for 30 min.
Determine frequencies present in record.
Percentage of time represented by each frequency.
All studies with ECS give score of zero.
Amplitudes 2–10 μV scored ×10 of amp., max. 100.
For other records, amplitudes disregarded.
Mean frequency of any range multiplied by incidence for frequency range.

Synek (1988)

Grade 1: predominantly normal alpha, some scattered theta.
Grade 2: dominant activity in theta range.
Subgrades:
A: widespread rhythmic theta, immediate attenuation by external stimulation.
B: similar pattern, no response to stimulation.
Grade 3: dominant delta activity.
Subgrades:
A: large amplitude rhythmic delta activity, maximal frontally.
B: 'preserved sleep spindles like stage II.'
C: diffuse, small amplitude delta.
D: moderate amplitude delta.
Grade 4: A: burst suppression.
B: polyspike or clusters of spikes.
C: alpha pattern coma.
D: theta pattern coma.
E: low output EEG.
Grade 5: isoelectric EEG.

TABLE III

Demographic data of trauma patients.

N	57 patients
Age	37.5 ± 2.8 years (4–81)
M:F	38 m : 19 f
Mechanism of injury	
MVA	47
Fall	7
Skiing	1
Gunshot wound	1
Crush	1
Time (days) injury to first EEG	9.9 ± 12.6 days
Medications (time of study)	
Paralyzing agents	9/57
Morphine/narcotics	26/57
Benzodiazepines	25/57
Anticonvulsants	9/57
Major tranquilizers	3/57
CT scans	
Abnormal	36/57
Findings	
SAH	14/57
Contusions	13/57
Intraventricular hemorrhage	8/57
Subdural	6/57
Epidural	3/57
Intracerebral hemorrhage	4/57
Ischemic stroke	2/57
Outcome (discharge from acute care hospital)	
Good	12
Moderate disability	8
Severe disability	17
Vegetative state	6
Death	14

All records were scored using the Hockaday, Hughes and Synek scales as well as the dichotomous scales. All studies could be scored using the dichotomous scale. Using the first reading (ARG) 7/57 records were not scorable using the Hockaday, 2/57 using the Hughes, and 2/57 using the Synek scales. These were due primarily to EEGs which did not fit into the definition of any of the categories, or fell into more than one category. For example, 2 patients with records dominated by focal status with polyspike could not be rated on the Hockaday score. A burst suppression pattern was difficult to rate using the Hughes score because it was impossible to assess the frequencies present. Patients with mixtures of theta and delta were difficult to place in the Synek score.

There were significant correlations between the previously described rating scales of Hockaday et al. (1965) and Synek (1988a,b) and the dichotomous scale (Table IV). An inverse correlation between the Synek and the

TABLE IV

Correlation between the dichotomous rating scale and Hockaday, Hughes, and Synek rating scales.

	Correlation matrix		
	Dichotomous	Hockaday	Synek
Hockaday	−0.806 *		
Synek	−0.738 *	0.787 *	
Hughes	0.369 *	−0.327 **	−0.08

* $P < 0.0001$; ** $P < 0.05$.

dichotomous scale was stronger than that seen with the Hockaday scale. All correlations were statistically significant at the $P < 0.001$ level. The Hughes scale did not correlate well with the other 3 scales.

A comparison between the first reading and second reading for one electroencephalographer (ARG) had a Pearson correlation coefficient of 0.95 (2-tailed significance < 0.001) (Table V). Inter-rater correlations were also highly significant. There was a 0.85 r correlation between readers ARG and PB, with similar correlation using the Hockaday and Synek scales. Because of the difficulties using the Hughes scale (subjective rating of frequencies) these were not rated by the second reader and no comparisons were available.

In this study 4 patients had EEGs consistent with ECI, and 13 patients had EEGs with reactive alpha (i.e., dichotomous score greater that 15, Hockaday score of 1, Synek score of 1). To ensure that the significant association was not due to the presence of EEGs at the "ends of the spectrum" we reanalyzed our data excluding these studies. We restricted the evaluation to patients with Hockaday scores of 2–4, Synek scores of major grades 2–4 and dichotomous grades of 15 to −9. Forty-one of the patients met these exclusionary criteria. Even without the presence of the extreme scores, the correlation between inter-rater and intra-rater readings was high (see Table 5). Again this correlation was best for the dichotomous scale. Data for the intra and inter-rater correlations on this group are plotted in Figs. 1 and 2.

A comparison between Glasgow outcome score at discharge and first dichotomous EEG score is illustrated in Fig. 3. There was a significant correlation between the EEG scale and GOS at the $P < 0.0001$ significance level. Patients with first EEG consistent with brain death were excluded from this analysis.

TABLE V

Intra- and inter-rater testing of the dichotomous scale.

(A) All patients, first EEG
(1) Intra-rater test/retest correlation $r = 0.9484$, $P < 0.001$, N = 57
(2) Inter-rater test/retest correlation $r = 0.8548$, $P < 0.001$, N = 57
(B) Excluding patients with scores of −10 or > 15
(1) Intra-rater test/retest correlation $r = 0.8082$, $P < 0.001$, N = 41
(2) Inter-rater test/retest correlation $r = 0.7655$, $P < 0.001$, N = 41

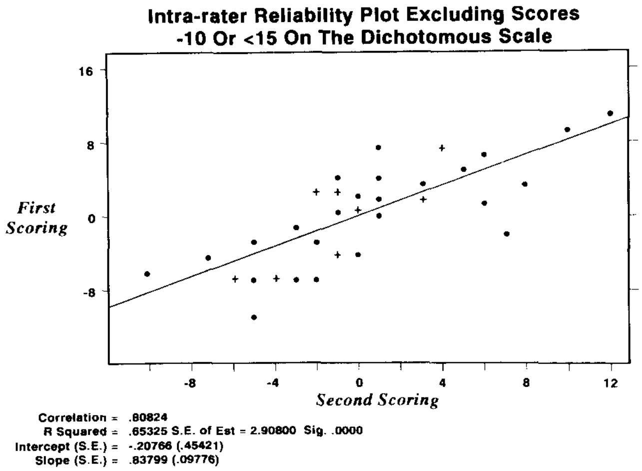

Fig. 1. Intra-rater reliability plot, first scoring – second scoring. ● = 1 observation; + = 2 observations.

The Synek scale also had good intra- and inter-observer reproducibility. This has not been previously shown. The scale's categorical nature limits its applicability in comparative studies. For the purpose of comparison in this study, we assigned fractional values to the subsets of the grades in the Synek scale to allow for statistical analysis. However, the subgrades often do not have similar prognostic value, limiting this process.

Other significant limitations of prior scales include the following:

(1) Hughes scale: with alpha coma or when two or more frequencies were present, the Hughes score was high, giving a false impression of a "good prognosis." Burst suppression and epileptiform activity were similarly difficult to score. Intra-observer reproducibility was low compared with the 3 other scales.

(2) Hockaday scale: certain subsets of coma such as alpha and theta coma patterns could not be scored.

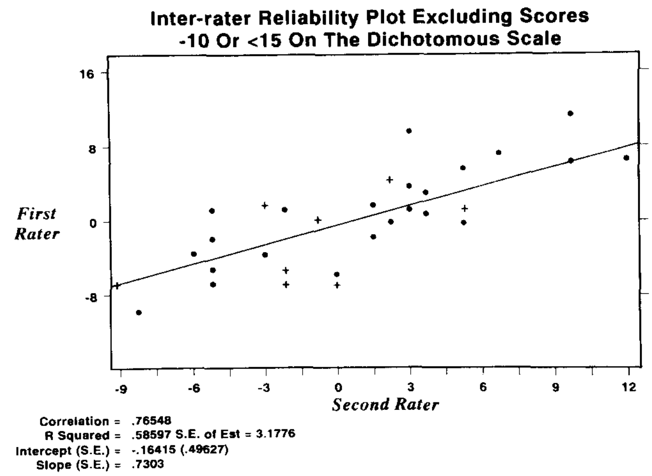

Fig. 2. Intra-rater reliability plot, first rater – second rater. ● = 1 observation; + = 2 observations.

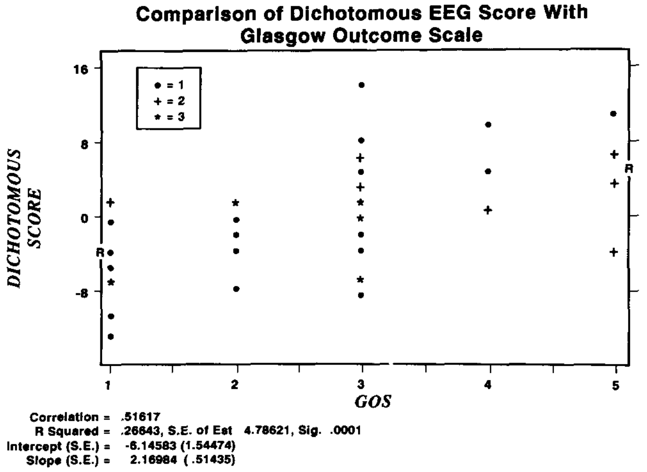

Fig. 3. Comparison of dichotomous EEG score with Glasgow outcome scale. GOS 1: death; 2: persistent vegetative state; 3: severe disability; 4: moderate disability; 5: good recovery (Jennett et al. 1976). ● = 1 observation; + = 2 observations, * = 3 observations.

Records with diphasic shifts from theta to delta patterns, and records with epileptiform activity were not scorable using this scale.

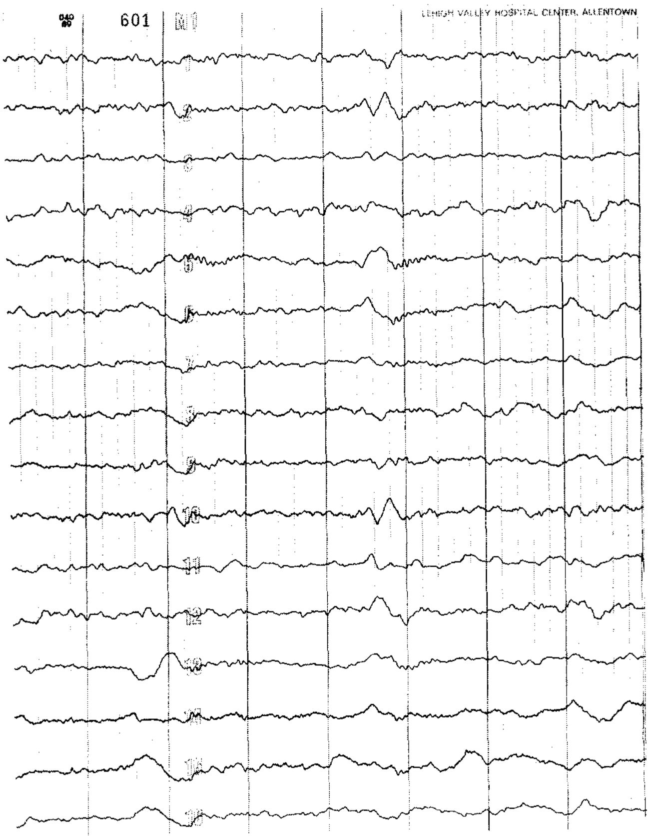

Fig. 4. Twenty-two-year-old male, comatose, 13 days post MVA. Full recovery. EEG with intermittent delta and theta, asymmetric spindles, reactive and variable, beta present. Dichotomous score 7. Sensitivity = 7 μV/mm, filters = 0.3–70 Hz, timebase = 1 sec/major division. Basic wave forms: Fp1-F3, F3-C3, C3-P3, P3-O1, Fp2-F4, F4-C4, C4-P4, P4-O2, Fp1-F7, F7-T3, T3-T5, T5–01, Fp2-F8, F8-T4, T4-T6, T6-O2.

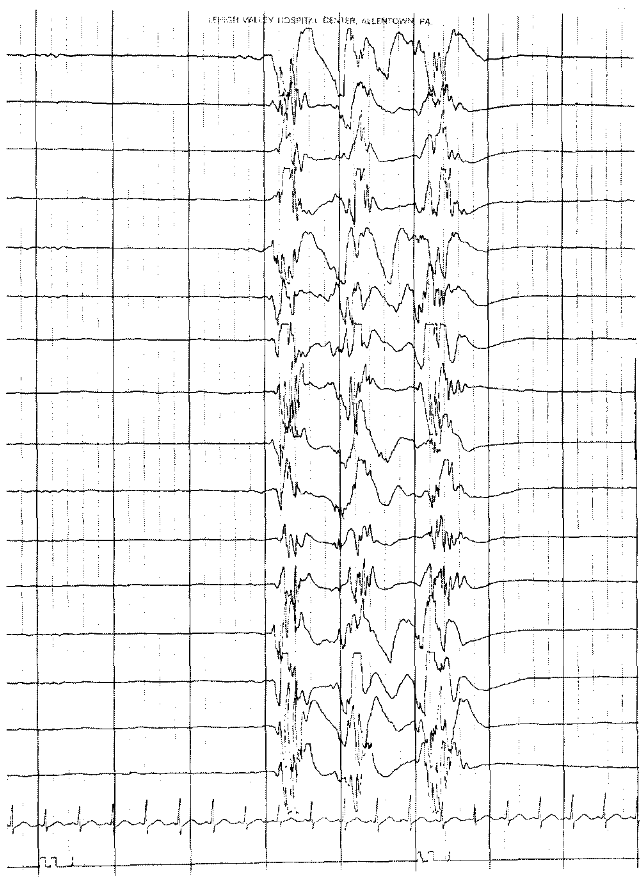

Fig. 5. Four-year-old male 1 day post MVA with head injury. Died day 3. Burst suppression pattern dichotomous score −5. Sensitivity = 10 μV/mm, filters = 0.3–70 Hz, timebase = 1 sec/major division. Basic wave forms: Fp1-F3, F3-C3, C3-P3, P3-O1, Fp2-F4, F4-C4, C4-P4, P4-O2, Fp1-F7, F7-T3, T3-T5, T5-O1, Fp2-F8, F8-T4, T4-T6, T6-O2.

(3) Synek scale: Synek scales were applicable to most cases. In some of our patients mixtures of theta and delta activity made differentiation between grade 2 and 3 difficult. This corresponded to the "Diphasic" group of Bricolo and Turella (1973). In a small number of cases identifiable spindles were seen but did not appear frequently during the record. Whether to score them as Synek's grade 3B or another grade on the basis of other background activity was unclear.

Representative examples of different EEG patterns are shown in Figs. 4–7. Examples of asymmetric spindle activity (Fig. 4), burst suppression (Fig. 5), posterior suppression (Fig. 6), and variable theta and delta (Fig. 7) are shown. A description of the basic wave forms present in each and the dichotomous score are presented in the caption of each figure. No specific changes attributed to medications were found in these patients. Serial EEGs, which were not available, might have made such changes apparent.

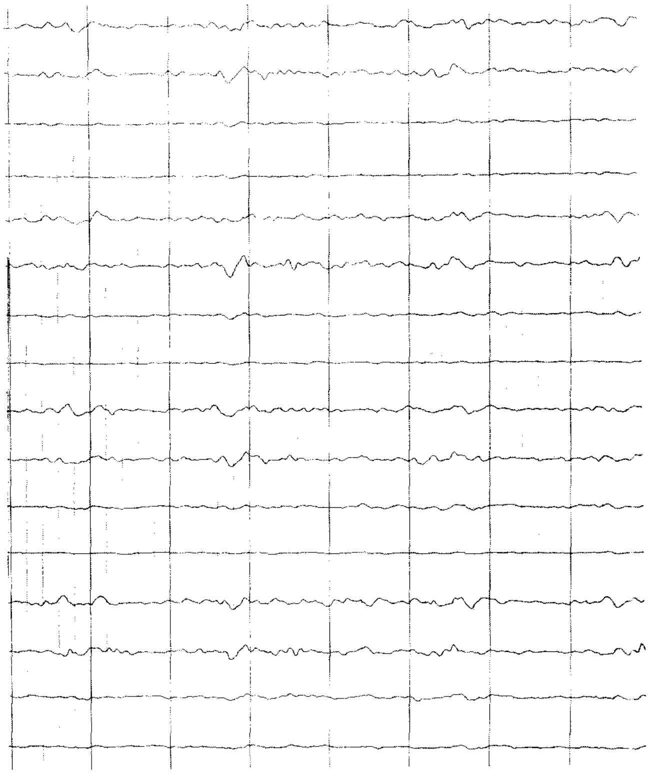

Fig. 6. Seventy-one-year-old female 1 day post MVA. Died day 8. Low voltage irregular theta and delta. Unreactive. Posterior suppression. Dichotomous score −2. Sensitivity = 5 μV/mm, filters = 0.3–70 Hz, timebase = 1 sec/major division. Basic wave forms: Fp1-F3, F3-C3, C3-P3, P3-O1, Fp2-F4, F4-C4, C4-P4, P4-O2, Fp1-F7, F7-T3, T3-T5, T5-O1, Fp2-F8, F8-T4, T4-T6, T6-O1.

Discussion

In certain conditions the EEG retains a pivotal role because it can disclose physiological abnormalities not appreciated with any other modality (Markand 1984). Spikes, electrographic seizures and focal slowing may all be present in a EEG while MRI, CT, vascular studies and evoked potentials are unremarkable (Engel 1989). In metabolic comas, where other studies of cerebral function are unremarkable and the CSF is clear, the EEG may be diagnostic (Markand 1984). In anoxic brain damage, a rich variety of EEG abnormalities are present (Bauer 1987).

In trauma, the presence of a clot, contusion or infarction are the "tip of the iceberg" in terms of the neurophysiological disruption which has occurred. Axonal shearing, seizure activity, and brain-stem disruption are poorly represented on most imaging techniques (Jennett and Teasdale 1981). The neurological examination is limited by the use of paralyzing agents, acute trauma triage, and the lack of good cortical discriminators in the physical examination once coma is present (Hall et al. 1985). EEG shows a rich field of changes in traumatic coma which serve as a marker for neurophysiological abnormalities.

Since the seminal works of Williams (1941a,b), Dow et al. (1944), Walker et al. (1944), and Dawson et al. (1951), a variety of EEG phenomena with trauma have been documented. Background abnormalities include electrocerebral inactivity (Courjon and Scherzer 1972), low voltage pattern (Synek 1988a,b), theta (Courjon and Scherzer 1972), delta including focal, diffuse, and intermittent rhythmic patterns (Stockard et al. 1975), fast 15–20 Hz activity (Loeb et al. 1959). Epileptiform patterns are frequent, and include focal spikes (Williams 1941b) and focal and generalized seizure activity, often with electrographic status (Silverman 1963; Synek 1988a,b). Periodic patterns include burst suppression (Zaret 1985), synchronous periodic slow wave bursts (Fischgold and Mathis 1959), and triphasic waves (Stockard et al. 1975). Focal abnormalities include focal flattening secondary to intracranial hematoma,

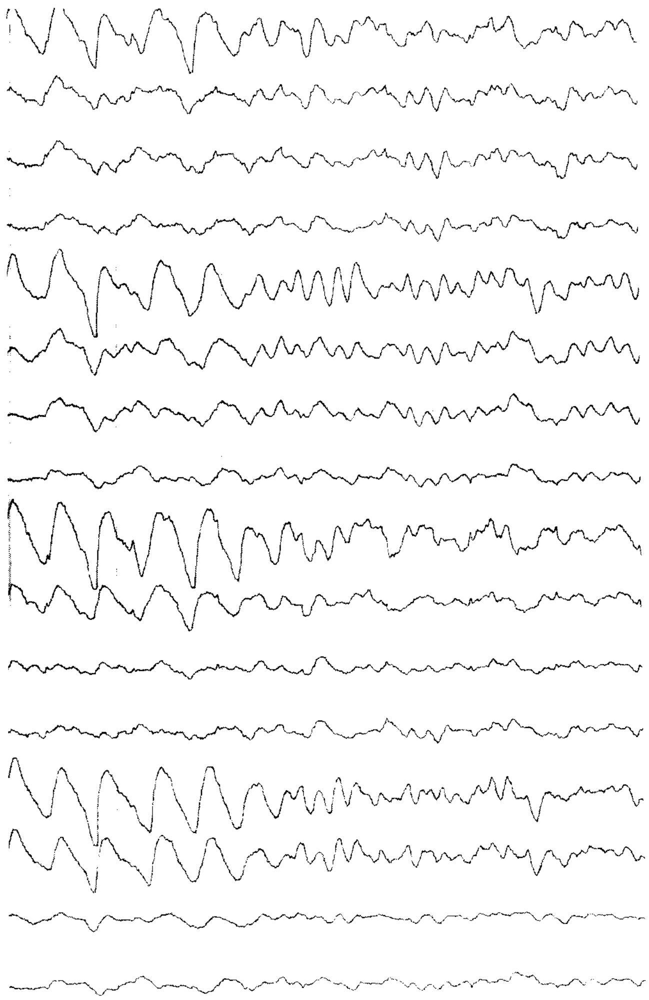

Fig. 7. Twenty-six-year-old female 4 days post MVA. Outcome severely disabled. Variable mixture delta and theta, unresponsive to pain, IRDA. Dichotomous score 3. Sensitivity = 7 μV/mm, filters = 0.3–70 Hz, timebase = 1 sec/major division. Basic wave forms: F3-A1, C3-A1, P3-A1, O1-A1, F4-A2, C4-A2, P4-A2, O2-A2, Fp1-A1, F7-A1, T3-A1, T5-A1, Fp2-A2, F8-A2, T4-A2, T6-A2.

edema, or infarction (Dow et al. 1944; Dawson et al. 1951), and asymmetry of background activity (Courjon and Scherzer 1972). Specific comatose patterns seen include alpha coma (Westmoreland et al. 1975), theta coma (Synek and Synek 1984, 1987) and spindle coma (Chatrian et al. 1963). The presence of variability and reactivity with different arousal patterns has been discussed (Dusser et al. 1989; Synek 1990). Circadian sleep wake cycles are also present in patients monitored over long periods (Chatrian et al. 1963; Bergamasco et al. 1968).

In most studies these phenomena have been isolated and correlated with outcome individually. Certain features have been considered to suggest a poor prognosis in trauma such as electrocerebral inactivity, "low voltage" or "low output" EEG, and burst suppression (Hughes et al. 1976; Kuroiwa and Celesia 1980; Sananman 1983). Burst suppression appears when anoxia complicates trauma (Stockard et al. 1975). Coma with invariant, widespread or anterior accentuated alpha has been associated with poor though not hopeless prognosis (Westmoreland et al. 1975; Austin et al. 1988). This pattern may occur in brain-stem trauma (Chatrian et al. 1964) and more usually with anoxic encephalopathy (Iragui and McCutcheon 1983; Austin et al. 1988).

Other features seem to bode well for prognosis. The presence of a variable and reactive background has both teleological and literature support as a good prognostic factor (Stockard et al. 1975; Markand 1984; Dusser et al. 1989; Synek 1990). A "diphasic" pattern with alternation from one background to another was considered a good prognostic feature by Bricolo and Turella 1973). This pattern corresponds to the variability subrating on the dichotomous scale. Spindles of normal morphology in many studies have been associated with a "good" resolution (Chatrian et al. 1963; Bergamasco et al. 1968; Bricolo and Turella 1973; Luecking 1979). In some papers, the presence of major asymmetries or "atypical" spindles was a poor prognostic feature (Rumpl et al. 1983). Some authors have also disputed the prognostic significance of spindles at all in these patients (Hughes et al. 1976; Hansotia et al. 1981). EEGs with a "monorhythmic" background were found to correlate with a poorer prognosis (Bricolo and Turella 1973; Dusser et al. 1989). An improvement in sequential EEGs has in early studies and in one study of children been shown to have utility in gauging outcome (Silverman 1963; Dusser et al. 1989).

Combining these independent EEG patterns into a numerical scale is possible. Using the dichotomous scale we were able to show that a high degree of intra- and inter-rater reproducibility was also possible. In addition, the scale is valid in terms of outcome judged retrospectively. Though visual interpretation of the EEG is subjective, using a standardized rating scale which minimizes subjective features of the electroencephalogram such a amplitude, percentage of activity, and subtle localizations, a high reproducibility can be achieved. In addition, using a weighting score to summate individual prognostic significance of EEG features, a valid assessment of prognosis is also achieved.

This is the first time to our knowledge that the EEG has been subjected to such a trial of reproducibility. Thus the dichotomous scale provides a more "objective" tool to further explore features of prognosis which may not have been initially apparent in the review of such traumatic EEGs. Because the subscales are scored and then weighted, a reweighting of variables is possible to improve the correlation with outcome variables. In addition, the dichotomous scale approximates a continuous scale (Hulley and Cummings 1988) and allows an effective statistical correlation with other clinical and neurophysiological scales. Could combinations of different tests improve prognostic scaling in subsets of coma? With a reliable and reproducible EEG tool, such a question may be answerable.

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